# Binary submasks and the middle column of Rules 126 and 129*

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**Abstract**

We formulate a target characterization of the rows of elementary cellular automaton Rule 126 from a single live cell, and its consequence for the middle column of Rule 129 under an exact complementation reduction. The candidate support is obtained by reflecting a union indexed by the binary submasks of the lower part of the generation number. For a generation with $p$ set bits, this gives $2^{p+1}$ live cells except at $n = 2^p - 1$, where the origin is counted once and the candidate row is full. The resulting middle-column identity is $\texttt{A267442}(n) = \texttt{A043545}(n)$ for $n \geq 1$, through `A036987`; the equality fails at $n = 0$. We report finite computational evidence for popcounts 1 through 18, including the second-smallest $p$-bit threshold, and specify a binomial-parity and Lucas-theorem roadmap. The universal row characterization and its Lean formalization remain pending. The sequence identity is already an OEIS conjecture, and related row-count and binomial formulas are prior art; no novelty is claimed for those assertions.

**Status and scope.** In the title footnote, "computational evidence complete" means that the supplied reports record completion of their finite checks. It does not mean exhaustive verification of the universal statement. "Proof pinned" is the drafting brief's project-status wording: the supplied reports still list pin-freezing as a next step, and contain no frozen Lean artifact. This paper neither certifies that status nor supplies the missing proof.

## 1 Introduction

Elementary cellular automata update a bi-infinite binary row using the same Boolean function of three neighboring cells at every position. In the standard numbering convention, Rule 126 returns zero on 000 and 111 and one on the remaining six triples. Rule 129 has the complementary output table. Wolfram's *A New Kind of Science* describes these automata and the relation of Rule 126 to Rule 90 [9]; it also discusses the power-of-two pattern in the center column of Rule 129 [10].

The sequence of center-cell states for Rule 129 from a single live cell is catalogued as OEIS A267442 [1]. The two comparison sequences are

$$r(n) := \texttt{A036987}(n) = \mathbf{1}_{\{n=2^j-1 \text{ for some } j\geq 0\}}, \tag{1}$$

$$q(n) := \texttt{A043545}(n) = 1 - r(n), \qquad n \geq 0. \tag{2}$$

The first is the Fredholm–Rueppel sequence; the second is its binary complement [2, 3]. The intended headline identity is

$$\boxed{\texttt{A267442}(n) = \texttt{A043545}(n) \quad (n \geq 1).} \tag{3}$$

*Preliminary draft — Lean formalization in progress; computational evidence complete, proof pinned and in preparation.

The positive-index qualification is essential. A267442 has value 1 at index zero, whereas A043545 has value 0 there. Richard J. Mathar's 17 January 2016 comment on A267442 already states (3) as a conjecture [1].

Our purpose is to organize a precise row-support statement, the finite evidence supplied for it, and the obligations of a subsequent formal derivation. The middle column is the simplest consequence of the proposed support formula. The threshold

$$T_p = 3 \cdot 2^{p-1} - 1$$

has an elementary interpretation: it is the second-smallest integer with $p$ set bits. Thus the apparent gap below the threshold contains only the single smaller $p$-bit integer $2^p - 1$, and no other candidate generations.

## 2 Conventions and the exact complementation reduction

Let $B_n, S_n : \mathbb{Z} \longrightarrow \{0, 1\}$ be the Rule 126 and Rule 129 rows, respectively, with the *same* initial condition

$$B_0(x) = S_0(x) = \mathbf{1}_{\{x=0\}}. \tag{4}$$

Writing $f_{126}, f_{129}$ for the two local maps, the updates are

$$B_{n+1}(x) = f_{126}(B_n(x-1), B_n(x), B_n(x+1)),$$

and likewise for $S$ with $f_{129}$. Explicitly,

$$f_{126}(a, b, c) = \mathbf{1}_{\{(a,b,c) \notin \{(0,0,0),(1,1,1)\}\}}, \qquad f_{129} = 1 - f_{126}.$$

Both maps are invariant under complementing all three inputs. If $C$ is pointwise complementation and $F_{126}, F_{129}$ are the global update maps, these truth-table identities give

$$F_{129} = CF_{126}, \qquad F_{126}C = F_{126}.$$

Starting at time one and iterating these equalities yields the exact reduction

$$S_n(x) = 1 - B_n(x) \qquad (n \geq 1,\ x \in \mathbb{Z}). \tag{5}$$

There is no alternating sign in time. At time zero, (4) applies instead.

The Rule 126 row vanishes outside the cone $|x| \leq n$ and is symmetric under $x \mapsto -x$. We write

$$C_B(n) = \#\{x \in \mathbb{Z} : B_n(x) = 1\}, \qquad C_S^{\text{cone}}(n) = \#\{x \in [-n, n] \cap \mathbb{Z} : S_n(x) = 1\}.$$

For $n \geq 1$, the Rule 129 background is one outside the cone, so a count of all its live cells would be infinite. The finite convention used here is

$$C_S^{\text{cone}}(n) = 2n + 1 - C_B(n) \qquad (n \geq 1). \tag{6}$$

This makes explicit the cone interpretation needed for the reports' Mersenne assertion `count_S` $= 0$; their symbol is not separately defined in the supplied packet.

# 3 The target row characterization and the middle column

For a positive integer $n$, define

$$p = \mathrm{popcount}(n), \qquad H(n) = 2^{\lfloor \log_2 n \rfloor}, \qquad L(n) = n - H(n).$$

Here popcount counts the one-bits in the binary expansion, $H$ is the leading power of two, and $L$ is the *lower part*. For nonnegative integers, $k \preceq L$ means that every set bit of $k$ is a set bit of $L$; equivalently, the bitwise intersection of $k$ and $L$ is $k$. The lower part has $p-1$ set bits and hence $2^{p-1}$ submasks. Define

$$U(n) = \bigcup_{k \preceq L(n)} \{n - 2k,\ n - 2k - 1\}. \tag{7}$$

This is a set of candidate *absolute positions*; reflection is required to obtain positions in the whole row.

**Target Theorem 1** (Lower-part support and Mersenne exception; pending proof)**.** *Let $p \geq 1$ and $n \geq 1$ be integers with* $\mathrm{popcount}(n) = p$*. If*

$$n \geq 3 \cdot 2^{p-1} - 1, \qquad n \neq 2^p - 1, \tag{8}$$

*then the proposed generation-$n$ Rule 126 row is characterized by*

$$B_n(x) = 1 \quad \Longleftrightarrow \quad |x| \in U(n) \qquad (x \in \mathbb{Z}). \tag{9}$$

*The union has $2^p$ distinct positive values, and*

$$C_B(n) = 2^{p+1}, \qquad B_n(0) = 0, \qquad S_n(0) = 1. \tag{10}$$

*At the exceptional generation $n = 2^p - 1$, the same union description gives*

$$U(n) = \{0, 1, \ldots, n\}, \qquad B_n(x) = \mathbf{1}_{\{|x| \leq n\}}, \qquad C_B(n) = 2^{p+1} - 1. \tag{11}$$

*The value zero is produced exactly once in* (7)*, from $k = L(n)$ in the term $n - 2k - 1$. Consequently $B_n(0) = 1$, $S_n(0) = 0$, and $C_S^{\mathrm{cone}}(n) = 0$.*

**Logical status.** Target Theorem 1 states the intended universal result, for arbitrary $p \geq 1$. It is not established here by the reported finite checks. Those checks concern specified samples in the tiers $1 \leq p \leq 18$; they do not quantify over every generation of even those tiers. The redundant exclusion in (8) is retained to match the consolidated statement supplied for this draft.

## 3.1 Why the threshold is the second $p$-bit integer

The smallest integer with $p$ set bits is $M_p = 2^p - 1$, using positions $0, \ldots, p-1$. Any other integer with $p$ set bits must use a position at least $p$. Its other $p-1$ bits contribute at least $2^{p-1} - 1$. Equality is attained by using positions $0, \ldots, p-2, p$, giving

$$T_p = 2^p + (2^{p-1} - 1) = 3 \cdot 2^{p-1} - 1.$$

For $p = 1$ the lower list of positions is empty, so the same reasoning gives $M_1 = 1$ and $T_1 = 2$. In particular, among integers of popcount $p$,

$$n \geq T_p \quad \Longleftrightarrow \quad n \neq M_p. \tag{12}$$

This is an ordering argument about binary integers, independent of any cellular-automaton assertion. There is no interval of additional $p$-bit test cases between $M_p$ and $T_p$.

### 3.2 The conditional middle-column consequence

Once Target Theorem 1, including its exceptional case, is established, its center-cell consequence is

$$B_n(0) = r(n) \quad (n \geq 0), \qquad S_n(0) = q(n) \quad (n \geq 1). \tag{13}$$

The first equality at $n = 0$ follows separately from the seed. Equations (5) and (1) then give the intended positive-index identity (3). The endpoint distinction is visible in the sequence values:

| $n$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| A036987 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| A043545 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 1 |
| A267442 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 1 |

These initial values illustrate the indexing; a finite matching prefix does not settle (3).

## 4 Proof roadmap: obligations and established ingredients

This section is a roadmap, not a proof of Target Theorem 1. The distinction is between the unresolved connection to the nonlinear evolution and the subsequent binary algebra.

**Step 1: fix the automata and the reduction.** Formalize (4), the two truth tables, finite propagation, reflection symmetry, and (5). The Boolean reduction in Section 2 is elementary; its Lean implementation and the positive-time domain still have to be linked to the eventual statement.

**Step 2: connect the evolution to binomial parity (pending).** A candidate closed form consistent with the lower-part union is

$$\widehat{B}_n(x) = \begin{cases} \binom{n}{\lfloor (n - |x|)/2 \rfloor} \bmod 2, & |x| \leq n, \\ 0, & |x| > n. \end{cases} \tag{14}$$

The upper binomial index is $n$. The key missing obligation is $B_n(x) = \widehat{B}_n(x)$ for all $n$ and $x$, including the cone edges, the center, both spatial signs, and the seed. One possible route uses Rule 90 or a suitable auxiliary recurrence, followed by Pascal's identity modulo two. Rule 126 itself is nonlinear, so its update cannot simply be replaced by XOR. This candidate is an explicit roadmap formulation; the two supplied reports do not contain a frozen statement or a derivation of (14). A related binomial formula for the right half of the Rule 126 triangle is already catalogued as A127872 [6].

**Step 3: apply the established Lucas criterion.** Lucas's theorem modulo a prime, here specialized to the prime 2, gives

$$\binom{n}{k} \equiv \prod_{i \geq 0} \binom{n_i}{k_i} \pmod 2, \qquad \binom{n}{k} \text{ odd} \iff k \preceq n, \tag{15}$$

where $n_i, k_i \in \{0, 1\}$ are binary digits [8]. For $n \geq 1$ and $|x| \leq n$, set $k = \lfloor (n - |x|)/2 \rfloor$. Then

$$0 \leq k \leq \lfloor n/2 \rfloor < H(n).$$

Thus $k \preceq n$ is equivalent to $k \preceq L(n)$. The floor equation is in turn equivalent to $|x| = n - 2k$ or $|x| = n - 2k - 1$. Accordingly Lucas's theorem converts (14) into the union (7). This conversion does not discharge Step 2.

**Step 4: count the candidate union.** Within either family $n - 2k$ or $n - 2k - 1$, different submasks give different values; between the two families, parity differs. Hence $U(n)$ has $2^p$ distinct values. Its minimum is

$$\min U(n) = n - 2L(n) - 1 = H(n) - L(n) - 1 \geq 0. \tag{16}$$

It is zero exactly when $L(n) = H(n) - 1$, or $n$ is Mersenne. Every positive absolute position yields two reflected cells, while zero yields one. These set-theoretic facts explain the two candidate counts and the center criterion. They are facts about $U(n)$; they apply to the automaton after the missing support equality is supplied.

**Step 5: complete the formal bridge.** The Lean development must identify its evolution, generation numbering, binary-popcount and submask definitions, integer coordinates, and natural binomial indices with the conventions above. It must then derive the exceptional case, the all-popcount assertion, and the positive-index OEIS identity from the actual row theorem. No Lean source, theorem identifier, dependency audit, or successful Lean build is part of the present evidence packet. The numbered target and equations here are editorial references, not claims that corresponding Lean declarations already exist.

# 5 Computational evidence and its exact limits

## 5.1 Provenance of the supplied evidence

The two supplied reports are dated 27 October 2026. The consolidation report [11] records a review of tranche output files, not a new calculation. For the power-of-two tier and popcounts 2 through 18, it records the terminal markers `ALL CHECKS PASS` and `TOTAL failures: 0`. The popcount-18 output has both markers twice: once for sections (A)–(E), and once for the (F) bitset run. The supporting `mersenne-rows`, `lucas-submask`, `diagonals`, and `rowcounts` tranches also carry `ALL CHECKS PASS`. The `a1a4-truthtables`, `a3lucas`, `closedform`, and `recurrence` tranches use different reporting conventions and are not cited there as tier evidence.

The threshold report [12] describes a separate `compute.py` run using Python and NumPy, with exit status zero and the same success markers. It states that no Lean was invoked. The present draft uses these two reports as evidence summaries; their underlying scripts and raw tranche logs were not supplied with this writing task.

## 5.2 All 18 tiers, without extrapolated ranges

For every tier $1 \leq p \leq 18$, the consolidation reports agreement with (7) and the non-Mersenne count on the generations checked in that tranche, as well as the exceptional full row. It also reports the converse test

$$C_B(n) = 2^{p+1} \quad \Longleftrightarrow \quad (\text{popcount}(n) = p \text{ and } n \neq 2^p - 1) \tag{17}$$

on each tranche's recorded converse interval. Only the largest such interval is reproduced: $p = 18$, $2 \leq n \leq 512000$. The other intervals are not specified. In particular, the upper endpoint 512000 must not be applied to every tier.

Table 1 reproduces every threshold and explicitly separates these unknown ranges. The count column evaluates the reported formula $2^{p+1}$; it is not a sample count. "Not supplied" means that the two input reports omit the relevant range, not that the original tranche performed no such check.

Table 1: Reported popcount tiers and bounds present in the source packet.

| $p$ | $T_p$ | $2^{p+1}$ | Analytic row sample | Converse interval | Simulation range |
|---|---|---|---|---|---|
| 1 | 2 | 4 | *not supplied* | *not supplied* | *not supplied* |
| 2 | 5 | 8 | *not supplied* | *not supplied* | *not supplied* |
| 3 | 11 | 16 | *not supplied* | *not supplied* | *not supplied* |
| 4 | 23 | 32 | *not supplied* | *not supplied* | *not supplied* |
| 5 | 47 | 64 | *not supplied* | *not supplied* | *not supplied* |
| 6 | 95 | 128 | *not supplied* | *not supplied* | *not supplied* |
| 7 | 191 | 256 | *not supplied* | *not supplied* | *not supplied* |
| 8 | 383 | 512 | *not supplied* | *not supplied* | *not supplied* |
| 9 | 767 | 1024 | *not supplied* | *not supplied* | *not supplied* |
| 10 | 1535 | 2048 | *not supplied* | *not supplied* | *not supplied* |
| 11 | 3071 | 4096 | *not supplied* | *not supplied* | *not supplied* |
| 12 | 6143 | 8192 | *not supplied* | *not supplied* | *not supplied* |
| 13 | 12287 | 16384 | *not supplied* | *not supplied* | *not supplied* |
| 14 | 24575 | 32768 | *not supplied* | *not supplied* | *not supplied* |
| 15 | 49151 | 65536 | *not supplied* | *not supplied* | *not supplied* |
| 16 | 98303 | 131072 | *not supplied* | *not supplied* | *not supplied* |
| 17 | 196607 | 262144 | *not supplied* | *not supplied* | *not supplied* |
| 18 | 393215 | 524288 | 19 values; see below | $2 \le n \le 512000$ | *not supplied* |

The threshold report identifies the $p = 18$ generation list with the popcount-18 tranche's list: the 19 integers having 18 set bits among positions $0, \dots, 18$. Equivalently this explicitly specified sample is

$$E_{18} = \{524287 - 2^j : 0 \le j \le 18\}. \tag{18}$$

The report describes these as 19 values $\le 524288$. This is a sparse list, not every generation up to that bound; its exact endpoints, from (18), are 262143 and 524286. The supplied packet does not identify the algorithm or coverage of every individual test on that list. In particular it does not give a bitset-simulation interval, even though it records completion of a bitset section. No analogous analytic row lists for the other 17 main tranches are reproduced.

### 5.3 Threshold-sharpness tranche: S1–S4

The separate threshold checks have a completely specified enumeration: for each $p = 1, \dots, 18$, the $p + 1$ values

$$E_p = \{2^{p+1} - 1 - 2^j : 0 \le j \le p\} \tag{19}$$

are all the integers with $p$ set bits in positions $0, \dots, p$. They exhaust the integers of popcount $p$ with $n < 2^{p+1}$, rather than all integers in that interval. A higher bit would already exceed the threshold. The report records the following four checks:

**S1.** At $n = T_p$, for every $p = 1, \ldots, 18$, the candidate union has $2^p$ distinct values, no collisions, and minimum $2^{p-1}$. The symmetry count is $2^{p+1}$. The report also specifies a pinned `math.comb` comparison for $n \leq 1000$. This is a cutoff for that comparison, not a declaration that every row up to 1000 was simulated. Among these threshold generations it includes $p \leq 9$ ($T_9 = 767$), and excludes $p \geq 10$ ($T_{10} = 1535$).

**S2.** Each complete list $E_p$ contains exactly two values at most $T_p$, namely $2^p - 1$ and $T_p$, and no value strictly between them. The $p = 18$ cross-consistency check is the 19-value list (18).

**S3.** At each of the 18 Mersenne generations $n = 2^p - 1$, the union has $2^p$ distinct values, with zero produced once at $k = L(n)$. The recorded symmetry count is $2^{p+1} - 1$.

**S4.** At every $n \in E_p$, for every $p = 1, \ldots, 18$, the minimum produced value is $n - 2k_{\max} - 1$, with $k_{\max} = L(n)$.

The report states that an earlier expectation of additional failing generations below $T_p$ was refuted by the enumeration: the proposed gap has no $p$-bit integers. The finding concerns ordering and the candidate union; it is not a new failure mode for the automaton.

In particular, the endpoint union properties were verified computationally for $n = T_p$ and $n = 2^p - 1$, $p = 1, \ldots, 18$, as reported in S1 and S3. This statement has those finite domains only.

The threshold tranche supplies no additional coverage assertion for $p > 18$ or for arbitrary generations above $T_p$. The elementary ordering argument in (12) is logically distinct from its finite enumeration checks. Likewise, direct counting of $U(n)$ is distinct from simulating Rule 126 or establishing the missing equality $B_n = \widehat{B}_n$.

## 6 Prior art and the limits of the novelty case

The submask/Lucas mechanism is folklore-grade. Lucas's theorem supplies the parity criterion, and the binary factorization of Pascal's triangle and its relation to additive cellular automata are standard ingredients. This draft claims no novelty for those ingredients.

There is also specific prior art beyond that general mechanism:

- A267442 already records the positive-index identity (3) as Mathar's 2016 conjecture [1]. Wolfram discusses the Rule 129 center-column pattern in 2002 [10].
- The Rule 126 triangle A071035 already identifies A036987 as its central column [4].
- A071051 already gives the Rule 126 row count $2^{1+\mathrm{popcount}(n)}$, with one subtracted at Mersenne generations, in a comment by N. J. A. Sloane dated 9 August 2014 [5]. Sloane also states this formula in Section 6.1 of his 2015 paper [7].
- A071035 identifies the triangle A127872 as the right half of its Rule 126 array [4]. A127872 includes G. C. Greubel's 18 April 2017 code for the formula

$$T(n,k) = \left( \binom{2n-1}{n-k} - \binom{2n-1}{n-k-2} \right) \bmod 2 \quad (0 \leq k \leq n,\ n > 0),$$

  with $T(0,0) = 1$ and out-of-range binomial coefficients set to zero [6]. This is prior art for a binomial representation, even though the entry does not supply the needed dynamical derivation.

The last formula is already algebraically equivalent to the candidate (14) on the right half. Put $m = n - k$. Two applications of Pascal's identity, followed by binary Lucas, give

$$\binom{2n-1}{m} - \binom{2n-1}{m-2} \equiv \binom{2n+1}{m} \equiv \binom{n}{\lfloor m/2 \rfloor} \pmod 2. \tag{20}$$

The first congruence uses that the middle Pascal coefficient is even and that plus and minus agree modulo two. Thus the algebraic comparison is available already; the remaining obligation is the dynamical derivation and its formalization, rather than this rewriting.

The precise lower-part union formulation (7) appears unpublished in the limited sources inspected for this draft. This is a provisional observation about presentation, not a priority claim: it is an elementary reformulation of the catalogued binomial expression, conditional on that expression's identification with the automaton. The identity itself is already published as a conjecture, so the stronger wording that both the characterization and identity are unpublished would be inaccurate. A completed machine-checked derivation may be a useful contribution, but mathematical novelty is not established by this draft or by the finite evidence. The limited source check did not review all linked manuscripts, including A. J. Macfarlane's manuscript linked from A071051 [5].

## 7 Work required before a proof-bearing preprint

The next mathematical deliverable is the missing recurrence-to-closed-form argument, followed by a Lean statement and checked derivation with the same seeds and indexing as this paper. The raw tranche reports and scripts are needed to replace the missing entries in Table 1 and to document which computations independently check the dynamics. Further literature review is needed before any novelty claim is finalized; the equivalence (20) already limits such a claim. Until those items are supplied, this manuscript is an evidence-based draft with a target theorem and a proof roadmap.

## References

[1] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A267442, central column of the single-seed Rule 129 evolution. Includes R. J. Mathar's conjecture, 17 January 2016. `https://oeis.org/A267442`.

[2] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A036987, Fredholm–Rueppel sequence. `https://oeis.org/A036987`.

[3] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A043545, complement of A036987. `https://oeis.org/A043545`.

[4] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A071035, Rule 126 triangle; central-column cross-reference. `https://oeis.org/A071035`.

[5] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A071051, Rule 126 row counts; formula comment by N. J. A. Sloane, 9 August 2014. `https://oeis.org/A071051`.

[6] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A127872, right half of the Rule 126 triangle. `https://oeis.org/A127872`.

[7] N. J. A. Sloane, *On the Number of ON Cells in Cellular Automata*, 2015, Section 6.1. arXiv:1503.01168v1. `https://arxiv.org/abs/1503.01168`.

[8] Édouard Lucas, *Sur les congruences des nombres eulériens et des coefficients différentiels des fonctions trigonométriques suivant un module premier*, Bulletin de la Société Mathématique de France **6** (1878), 49–54. doi:10.24033/bsmf.127. `https://www.numdam.org/item/BSMF_1878__6__49_1/`.

[9] Stephen Wolfram, *A New Kind of Science*, Wolfram Media, 2002, p. 55 (elementary rules) and pp. 269–270 (Rule 126 and Rule 90). `https://www.wolframscience.com/nks/p55--more-cellular-automata/`; `https://www.wolframscience.com/nks/p269--special-initial-conditions/`.

[10] Stephen Wolfram, *A New Kind of Science*, Wolfram Media, 2002, p. 641, discussion of the center column of Rule 129. `https://www.wolframscience.com/nks/p641--computations-in-cellular-automata/`.

[11] *Own-harness research—Rule 126 popcount series consolidation*, 27 October 2026, orchestrator run approximately 19:53 PDT. Unpublished supplied evidence summary, `REPORT-consolidation.md`.

[12] *Own-harness tranche—Rule 126 popcount threshold sharpness*, 27 October 2026, run approximately 20:23 PDT. Unpublished supplied evidence summary, `REPORT-threshold.md`.

Online references consulted 28 October 2026. The reports are local source documents, not public proof artifacts.